

Arpit Singh

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Pre-final year B.Tech student specializing in **ML, Deep Learning & AI** | Building towards **MLOps & CI/CD** | Interested in applying data-driven approaches to engineering and industrial systems | Seeking **Data Science / AI / ML** internships

EDUCATION

ABV-Indian Institute of Information Technology and Management (IIIT), Gwalior <i>B.Tech in Mathematics & Scientific Computing</i>	Aug 2023 – Aug 2027 <i>CGPA (5th Sem): 8.25</i>
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EXPERIENCE

Research Intern <i>ABV-IIITM Gwalior (Under Dr. Jeevaraj S)</i>	Dec 2025 – Feb 2026
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- Developed a **hybrid recommendation system** combining neural collaborative filtering (NCF) with uncertainty-aware matrix factorization.
- Implemented and optimized models using **Optuna**, achieving **HR@10 of 0.85** and **NDCG@10 of 0.58** on MovieLens datasets.
- Designed an **ensemble scoring framework** fusing multiple signals, outperforming NCF and MF baselines; evaluated across **ML-100K, 1M, and 10M** datasets. *Manuscript in preparation.*

PROJECTS

ReceiptSense <i>Tools: Python, FastAPI, PyTorch, Donut (CORD v2), React, Node.js, MongoDB</i>	Jan 2026 – Mar 2026
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- Developed an **AI-powered receipt analysis system** integrating a FastAPI ML microservice with a MERN stack for automated extraction and tracking of user expenditures.
- Implemented an end-to-end **document understanding pipeline** (image preprocessing → Donut inference → structured parsing → MongoDB) converting unstructured receipt images into analyzable financial data.
- Experimented with **fine-tuning the Donut model** on SROIE; selected CORD-v2 for superior real-world generalization. Designed analytics for item-level parsing, category classification, and spending trends.

ML-Based Inventory Optimization (MOIC Framework) <i>Tools: Python, XGBoost, CatBoost, Scikit-learn</i>	Sep 2025 – Nov 2025
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- Built a large-scale **demand forecasting and inventory optimization system** on a **2.7M+ sample** dataset, modeling real-world retail demand patterns across multiple categories. Performed **data preprocessing, feature engineering, and demand pattern classification** (Smooth, Intermittent, Lumpy, Erratic).
- Benchmarked **gradient boosting models (XGBoost, CatBoost)** in a MOIC framework; achieved **normalized cost of 0.92** and **99.4% service level**, outperforming FSH baselines (cost: **21.3→0.92**; service: **51.6%→99.4%**).
- Designed a **decision-oriented optimization framework** aligning predictive modeling with operational cost efficiency, also demonstrating applicability to supply chain and industrial optimization problems.

ACHIEVEMENTS

- Secured **98.45 percentile** in JEE Mains 2023, achieving All India Rank of **18,051** among 1M+ candidates.
- Awarded **NTSE Scholar** (2020) for outstanding academic performance at the national level.
- Qualified **GATE** in both CS/IT and Data Science & AI streams.

TECHNICAL SKILLS

- Languages:** Python, C++, C, Bash, SQL
- Machine Learning & AI:** Deep Learning (CNNs, TCNs), NLP, Large Language Models (LLMs), Generative AI (Transformers, Hugging Face) Time-Series Analysis, Recommendation Systems, Optimization, Probabilistic Modeling
- Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV, FastAPI
- Data & Visualization:** NumPy, Pandas, SciPy, Matplotlib **Tools:** Git, Docker (basic), REST APIs, Kaggle, Optuna | *Learning:* MLOps, CI/CD
- Coursework:** DSA, Operating Systems, Statistics, DBMS, Computer Networks, Numerical Analysis, Multivariate Data Analysis, Trustworthy AI

LEADERSHIP AND EXTRACURRICULAR

Executive, Manchtantra <i>ABV-IIITM Gwalior</i>	Mar 2025 – Present
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- Coordinating and overseeing all major club activities, including directing and mentoring a team of performers for national-level inter-college fests, coordinating rehearsals and creative execution to secure podium finishes in competitive drama categories

Content Team, IEEE <i>ABV-IIITM Gwalior</i>	Aug 2024 – Nov 2024
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- Managed event promotions and digital content strategy for IEEE initiatives, increasing audience outreach and event participation through targeted communication.